RAGHUVEER THIRUKOVALLURU

E-mail: raghuth@iitk.ac.in | Mobile No: + 91 965 142 4499

EDUCATION

Year	Degree/Certificate	Institution	CPI / %
2012-2016	B-Tech Electrical Engineering	IIT Kanpur	8.9/ 10
2012	Class XII State Board, Andhra Pradesh	Sri Chaitanya Junior Kalasala	94.3
2010	Class X State Board, Andhra Pradesh	St. Andrews High School	90

SCHOLASTIC ACCOMPLISHMENTS

- Secured an All India Rank of **26** in Kishore Vaigyanik Protsahan Yojana **(KVPY 2012)**
- Placed in **National top 1% at NSEP (2011-2012), National top 1% at NSEA (2009-2010)** by IAPT
- Secured **20th rank in STSE 2007, 49th rank in STSE 2008** conducted by **A S Rao Awards council**
- Secured an All India Rank of **588** (General Category) in **IIT-JEE 2012** among over 500,000 students
- Secured an All India Rank of **87**(General category) in B.A. **AIEEE 12** among over 600,000 students
- Secured a state rank of **256** (General Category) in **EAMCET-2012** conducted in Andhra Pradesh among 400,000 students
- Received **Notional Award** 2012-2013 for excellence in academics – awarded to top 7% students in our batch
- Placed **overall third** in TA202 among 70+ projects and **sectional third** in TA201 among 16+projects.

PROJECTS & DEVELOPMENT WORK

Local Transport Automation System : -*Course Project under Prof. T.V. Prabhakar, CSE, IITK, [link]* *(Jan'16-May'16)*

- A Bluetooth mobile network system which works **without establishing connection** to purchase tickets, transmit stop info, etc.
- Designed a dynamic **temporal architecture** alternating between info trans. to monetary transc. handling **> 200** indviduals on it.
- Other features include secure (AES 128 enc + Rijndael cipher) trans., Destination alarm, Transport wallet, Emergency stopper.
- **Runners up** to **Best Software Award(Pankaj Jalote Award)** amongst all projects (CSE, EE, IME) of the graduating batch of 2016.

Placement Portal: - *under Prof. Sumit Ganguly, CSE, IITK, [link]* *(Jan'16-May'16)*

- Designed a Institute placement portal for Admin, Companies and Students to handle incoming openings and applications
- Used a **MYSQL database** to store different interactions between entities and used a **PHP frontend** with an appealing UI
- Features include **advanced company search(based on field requirements, using keywaord & semantic search)** for students.
- Other features include admin approve messages, calendar for interviews, automatic deadline notifications etc.

MOBILE APPLICATIONS

Android Watch : - *under Electronic club, SnT Council, IITK, [Video]* *(May'13-July'13)*

- A Device which syncs with android, provides a way to control one's phone from his watch and organise information
- Used **Arduino Uno**, **RN-42 bluetooth** module and 4Display Shield 144 to create a cost efficient model-INR 5000
- Easy way to **transfer calls and messages, change profiles, set alarms, control music player, volume manager, gps, wifi**
- **Self designed the Communication Protocol,** designed an android app which handles data transfer and **manages functionality**
- Developed a system using Service class & Handlers enabling **background functionality** even when the phone is locked

SMSify : - *during Yahoo HackU and continuing* *(Aug'13-Sept'13)*

- This is a **two way SMS Gateway app for accessing public/private APIs** implemented on an android for the first time
- Designed a system using Worker threads and Async Tasks which can handle upto **simultaneous 128 tasks** on a single droid
- Developed a system where one can **dynamically create services for API access** of all kinds of json/xml/rss feeds
- Created a **JSONTREE UI** wherein the user could choose the values to be parsed from a **Structured Layer Tree** of the api
- Brings down **cost to 5% of the existing** services in India

Udghosh : - *under WebTeam Udghosh'14[link]* *(July'14-Oct'14)*

- This is the official android app of Udghosh-Annual Inter Collegiate Sports Festival of IIT Kanpur
- Designed an intra IITK **navigation system** with all the landmarks using **Google maps and Gps** in this app
- Implemented server-mobile notification feature in this app using **Google Cloud Messaging service**
- Currently working on **cache based storage** of images using **Picasso** and integration with **social networks**

Crisis Control : *(September'13)*

- This was an assistive app for the Uttarakhand flood victims which **won 2nd prize in Takneek'13**
- It Included a **Person finder, Resource finder** wherein people could report missing persons and scarce/excess of resources
- Used the **Facebook Graph API** to create **Story Sharer** through which one could post his experience onto a **fb page**
- Trauma Stabiliser was the other offline feature which helps in moral boosting & maintaining mental state of the the victims

FlirtAlert : - *during Microsoft code.fun.do* *(January'14)*

- This is a Dating service app developed on windows platform integrated with Facebook
- Used the **Facebook Graph API** and our server to match **impressions** of friends and connect them by push notifying them
- Used the **Windows Azure Service** for push notifications used to notify friends following impression match between them

Synchronicity : - *under Team Antaragni'13* *(October'13)*

- This was the official android app of Synchronicity-the annual rock music festival of Antaragni held at IIT Kanpur
- An informative app which gave information on history, past winners, prizes and sponsors

EMBEDDED SYSTEMS

Bubble Burster : *(September'13)*

- Game developed using microcontroller and a GU display wherein the player bursts bubbles on the screen using a keyboard
- **Won 4th in Takneek'13** among more than 15 participating teams for this effort
- Reviewed **keyboard protocols** and successfully encoded this in a microcontroller. Reviewed and **modified the microcontroller library** for a single keyboard to enable **simultaneous input from two keyboards**
- Highly appreciated for **animations** made up in bursting bubbles by creating three consecutive frames played continuously

Audio Recorder : *(March'14)*

- A Device which **amplifies** audio signals, **samples** them at a specified rate, converts them into **.wav** files and stores on SDcard
- Designed a **two order filter and amplifier** circuit for speech signal using **LM324 op amp** inputted using an electret mic
- Integrated two arduinos using **Serial Communication protocol** and used them to handle GLCD and SDcard respectively
- Used the Arduino **SdFat** library for storing recordings and **TMRpcm** for playback of the wave file stored on the SDcard

INTERNSHIPS

WINTER HACKATHON - *under Pulkit Agarwal (PhD U.C. Berkeley), Anubhav Singla (MS Stanford)[link]* *(Dec'13)*

- Got selected into **top 8 students among 100+ entries from all batches** in our institute
- Studied **Bluetooth Low Energy(BLE), Estimote** (BLE beacons) **android Sdk** and understood their communication protocols

INFEXIOUS
Android App developed

- A **Localized Social Network** app wherein infections can be spread over **BLE networks** without an established connection
- Implemented a Multithreaded system using AsyncTasks and Service class to enable **background functionality**
- Developed a system which **facilitated for offline detection** of devices and **dynamically update server** when online
- Designed a responsive UI to create, update, delete and store infections managed by a **SQlite** database

POSITIONS OF RESPONSIBILITY

Coordinator Tennis-Udghosh'13

- Lead an organizing team of 25 different judges, linesmen and ballboys during the event with a 20 team participation
- Contributed to smooth conduction despite random sudden withdrawal of teams, by dynamically rescheduling of matches
- Helped out managing a budget of over INR 30,000 during the event

Coordinator Duathalon-Udghosh'13

- Contributed in safe conduction of the marathon by helping out in aiding the hurt ones during the event
- Managed and helped out in efficient distribution of food and drinks over INR 5000

Member Web Team-Udghosh'14

- Introduced Result updates, Photo viewer, Map based navigation for the first time into its android app
- Introduced an image upload portal for the first time wherein photos of the festival could be viewed

Secretary Synchronicity-Antaragni'13

- A member of the calling team, called up many semiprofessional rock bands in the country

Team Mentor –Summer projects Electronics club'14

- Contributed to successful completion of projects of the mentees

RELEVANT COURSES

MACHINE LEARNING : *By Andrew NG, Stanford – (Scored 93%), Coursera* *(March'14-June'14)*

- Rigorously worked through the basics of supervised learning-multivariate linear, logistic regression with regularisation
- Studied and implemented a **nueral network for handwritten digit recognition** using one-vs-all logistic regression classifier
- Used **SVM** with gaussian kernels to build non linear classifiers and went on to build an **email spam classifier**
- Implemented **K-mean**s to achieve **image compression**; also built a **movie recommender** using **collaborative filtering**

OTHER COURSES : Data Structures & Algorithm | Probability and Statistics | Game Theory | Linear Algebra | Fund. of Computer Science |

TECHNICAL SKILLS

- Operating Systems: **Windows, Linux**
- Programming languages: **C & C++, Java and basics of Python, Javascript**
- IDE's: **Eclipse, Android Studio, Sublime text, Vim editor**
- Simulation Tools: **MATLAB, AutoCAD, MICROCAP, GNU OCTAVE**

EXTRA CURRICULAR ACTIVITIES

- Did well in programming contests like WPC 13' and ICPC –regionals, active problem solver on codechef.com
- Developed the snake game in Techkriti'13, and frog game in Takneek'12, active member of Electronic Club
- Highly interested in **Chess –A 1300 pointer** on (blitz, chess.com)
- Highly interested in **Tennis** and an active player in our campus